LECTURE 04 • DISCRETE MATHEMATICS

Equivalence laws, then predicate logic

Truth-preserving rewrites, open predicates, quantifiers, and negation

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BEFORE WE BEGIN

Announcements

01	LONG QUIZ 1	**Tue, Oct 6 • 6:00 PM • SDSB**
02	WORKSHEETS	**No devices during worksheet sessions.**
03	GRADING EMAILS	**Email Ahsan about grades, grading, or makeups.**
04	WORKSHEETS 1 AND 2	**Both worksheets have been graded.**

OPENING PUZZLE • THE SABOTAGED TELESCOPE

Exactly two reports are true

One of four astronomers sabotaged the telescope. Exactly two reports are true. Who did it?

ADA	**Basil sabotaged the telescope.**
BASIL	**Cyra did not sabotage the telescope.**
CYRA	**Dara sabotaged the telescope.**
DARA	**Ada is telling the truth.**
DARA	**Dara works. Basil and Cyra tell the truth. Ada and Dara do not. Every other suspect produces a different count.**

SO FAR

The logic toolkit so far

01	CLAIMS	**Propositions have one truth value in a fixed context.**
02	OPERATORS	**$\neg$, $\wedge$, $\vee$, $\to$, and $\leftrightarrow$ build compound claims.**
03	TESTS	**Truth tables and counterexamples expose invalid patterns.**
04	REWRITES	**A conditional matches its contrapositive and $\neg p \vee q$.**

Next, we organize these rewrites and extend the language to predicates.

TODAY

Equivalence laws and predicate logic

01

TEST

Do two formulas agree in every possible world?

02

REFACTOR

Replace a part with an equivalent form.

03

GENERALIZE

Turn repeated propositions into predicates.

04

NEGATE

Push NOT through quantifiers without losing meaning.

Where logic fits in the course

This lecture connects evaluating claims with using them to prove results.

p, q

Name claims

Propositions give each claim a truth value.

$p \equiv q$

Rewrite safely

Equivalence changes the form while preserving truth.

$P(x)$

Describe many objects

Predicates replace repeated claims with one open sentence.

$P_1, \ldots, P_n \vdash C$

Build proofs

Later, proof rules connect known facts to new conclusions.

The same language returns in proofs, sets, relations, functions, and graphs.

EXERCISE 1 • COMMIT BEFORE CALCULATING

Which formula is wearing the disguise?

Same claim, sometimes the same, or never the same?

$$\neg(p \lor (\neg p \land q))$$

vs

$$\neg p \land \neg q$$

SAME IN EVERY WORLD

SAME IN EVERY WORLD

SAME IN SOME WORLDS

NEVER THE SAME

Equivalent parts can be swapped inside a larger claim

$$\mathbf{r} \wedge (\mathbf{p} \rightarrow \mathbf{q}) \quad \rightleftarrows \quad \mathbf{r} \wedge (\neg \mathbf{p} \vee \mathbf{q})$$

The outside structure stays put. Only the equivalent component changes.

Prove useful moves once. Reuse them everywhere.

EXERCISE 2 • BUG HUNT

Four laws entered with forged passports

Circle every false equivalence. Repair it with the smallest possible edit.

AND + CONSTANTS

$p \wedge F \equiv p$

$p \wedge T \equiv p$

$p \wedge F \equiv F$ • domination

OR + CONSTANTS

$p \vee T \equiv T$

$p \vee F \equiv F$

$p \vee F \equiv p$ • identity

DOUBLE NEGATION

$\neg\neg p \equiv \neg p$

$\neg\neg p \equiv p$ • double negation

DE MORGAN

$\neg(p \wedge q) \equiv \neg p \wedge \neg q$

$\neg(p \wedge q) \equiv \neg p \vee \neg q$

$\neg(p \wedge q) \equiv \neg p \vee \neg q$ • De Morgan

Most laws perform one of six moves

SWAP	**Commutative** $p \lor q \equiv q \lor p$	REGROUP	**Associative** $(p \land q) \land r \equiv p \land (q \land r)$
DISTRIBUTE	**Distributive** $p \land (q \lor r) \equiv (p \land q) \lor (p \land r)$	SIMPLIFY	**Identity • domination • idempotent • complement** $p \lor F \equiv p$ • $p \land \neg p \equiv F$ • $p \lor \neg p \equiv T$
PUSH NOT	**De Morgan • double negation** $\neg(p \lor q) \equiv \neg p \land \neg q$	REMOVE ARROWS	**Conditional • biconditional** $p \to q \equiv \neg p \lor q$

IDENTITY • THE QUIET GUEST

True is neutral for AND; false is neutral for OR

$p \wedge T \equiv p$

TRUE adds no new restriction

$p \vee F \equiv p$

FALSE adds no new option

Identity changes the expression—not its information.

True swallows OR; false swallows AND

$p \vee T \equiv T$

one guaranteed route is enough

$p \wedge F \equiv F$

one impossible demand ruins everything

Some constants do not join the result. They take over.

IDEMPOTENT + DOUBLE NEGATION

Duplicates collapse; two NOTs cancel

$p \lor p \equiv p$

repeating an option adds nothing

$p \land p \equiv p$

repeating a demand adds nothing

$\neg\neg p \equiv p$

flip twice and return home

EXERCISE 3 • TRANSLATE THE NEGATION

The night shift denies the whole memo

Negate: "The server is up AND the backup is current."

$$\neg(a \land b) \equiv \neg a \lor \neg b$$

Either the server is not up, or the backup is not current.

CONDITIONAL NEGATION

To negate a promise, show it was made and broken

$$\neg(p \to q) \equiv p \land \neg q$$

p **the promise is triggered**

$\neg q$ **the promised outcome fails**

WORKED EXAMPLE • LABEL EVERY STEP

A long formula yields to one lawful move at a time

$\neg(p \vee (\neg p \wedge q))$

1	$\neg p \wedge \neg(\neg p \wedge q)$	De Morgan
2	$\neg p \wedge (p \vee \neg q)$	De Morgan + double negation
3	$(\neg p \wedge p) \vee (\neg p \wedge \neg q)$	Distributive
4	$F \vee (\neg p \wedge \neg q)$	Complement
5	$\neg p \wedge \neg q$	Identity

Simplification starts with a destination

1

NAME THE TARGET

What shape would be easier to use?

2

SCAN LOCALLY

Which familiar law appears right now?

3

WRITE ONE MOVE

Do not hide three laws on one line.

4

JUSTIFY IT

A correct answer without a route is fragile.

There may be many good routes. Every route needs receipts.

Refactor without building a truth table

Simplify $p \lor (\neg p \land q)$ without a truth table. Label every move.

$p \lor (\neg p \land q)$

1	$(p \lor \neg p) \land (p \lor q)$	Distributive
2	$T \land (p \lor q)$	Complement
3	$p \lor q$	Identity

Propositional logic runs out of name tags

Alice knows logic. Bob knows logic. Charlie knows logic… surely there is a shorter script.

L(x): x knows logic

A predicate keeps the sentence and opens a slot

L(x): x knows logic

A — **L(Alice)**
Alice knows logic

B — **L(Bob)**
Bob knows logic

C — **L(Charlie)**
Charlie knows logic

The variable is the empty name tag. The predicate is the reusable sentence.

Predicate logic has a cast, a script, and a stage

01	CONSTANT	**a specific object**	02	VARIABLE	**an open role**
	Earth · 2026 · Mona Lisa			x · y · z	
03	PREDICATE	**a property or relation**	04	DOMAIN	**who may audition**
	Prime(x) · Orbits(x, y)			integers · planets · students	

P(x): x IS PRIME

Plugging in an object closes the slot

P(x)

P(2)	P(3)	P(4)	P(5)	P(6)	P(7)
T	T	F	T	F	T
2 is prime	3 is prime	4 = 2 × 2	5 is prime	6 = 2 × 3	7 is prime

Domain: positive integers. P(x) is open; P(4) is a proposition with a truth value.

EXERCISE 5 • PREDICATE CASTING

Which triples earn a right-angle badge?

$Q(x, y, z): x^2 + y^2 = z^2$

Domain: positive integers. Decide before calculating every square.

Q(3, 4, 5)	Q(2, 2, 3)	Q(5, 12, 13)
$3^2 + 4^2 = 5^2$	$2^2 + 2^2 \neq 3^2$	$5^2 + 12^2 = 13^2$
T	F	T

CLOSE THE SLOT

Two machines turn a predicate into a proposition

SUBSTITUTE → $P(7)$	QUANTIFY → $\forall x\, P(x)$ or $\exists x\, P(x)$
plug in one object	speak about the domain

A value replaces the variable; a quantifier binds it. Either way, the formula closes.

For every is a giant AND

$\forall x\, P(x)$ $\equiv$ $P(a) \land P(b) \land P(c)$

TRUE every object passes

FALSE one counterexample is enough

For the finite domain {a, b, c}

There exists is a giant OR

$$\exists x\, P(x) \equiv P(a) \lor P(b) \lor P(c)$$

TRUE one witness is enough

FALSE every object must fail

For the finite domain {a, b, c}

The domain can flip the verdict without touching the formula

$\forall x \, (x^2 > x)$

POSITIVE INTEGERS

FALSE

$x=1$ is a counterexample

INTEGERS GREATER THAN 1

TRUE

every allowed x satisfies it

Never judge a quantified statement before reading its universe.

An open variable leaves truth waiting backstage

$x + 2 = 5$	OPEN	x still needs a value
$3 + 2 = 5$	CLOSED	now the statement is true
$\forall x \in \mathbb{Z}\ (x + 2 > x)$	CLOSED	the quantifier binds x

A proposition has no free variable occurrences.

Proof and disproof trade jobs across the quantifiers

CLAIM	TO PROVE	TO DISPROVE
$\forall x\, P(x)$	show every x works	find one x that fails
$\exists x\, P(x)$	find one x that works	show every x fails

One object can save ∃ or destroy ∀.

EXERCISE 6 • ENGLISH → SYMBOLS

Translate the policy without changing who it targets

Every CS student must take discrete mathematics.

C(x)	D(x)	DOMAIN
x is a CS student	x must take discrete mathematics	all students

$$\forall x\,(C(x) \rightarrow D(x))$$

Do not accidentally declare every student a CS student.